Definition 0.0.1. Axiom of Choice
Let S_α $(\alpha \in \Lambda)$ be non-empty sets.
For $\mathcal{S} \overset{\text{def}}{=} \{S_\alpha \mid \alpha \in \Lambda\}$, there exists a **Choice Function**:

$$f : \mathcal{S} \to \bigcup_{\alpha \in \Lambda} S_\alpha \ \text{ s.t } \ f(S_\alpha) \in S_\alpha, \ \forall \alpha \in A$$

0.1 Topology and Topological Space

Definition 0.1.1. Let X be a set. The collection $\mathcal{T} \subseteq X$ is called a **Topology** on X if $\mathcal{T}$ satisfies the following:

A1. $\emptyset, X \in \mathcal{T}$.

A2. For any subset $S \subseteq \mathcal{T}$, $\bigcup_{U \in S} U \in \mathcal{T}$.

A3. For any finite subset $F \subseteq \mathcal{T}$, $\bigcap_{U \in F} U \in \mathcal{T}$.

The tuple $(X, \mathcal{T})$ is called a *Topological Space*.

0.2 Basis

Definition 0.2.1. Let $(X, \mathcal{T})$ be a Topological Space. A subset $\beta \subseteq \mathcal{T}$ is called a **basis** of $\mathcal{T}$ if:

For any $U \in \mathcal{T}$, there exists $\beta^* \subseteq \beta$ such that $U = \bigcup_{B \in \beta^*} B$.

Theorem 0.2.2. Let $(X, \mathcal{T})$ be a topological space, and $\mathcal{C} \subseteq \mathcal{T}$. TFAE:

a) $\mathcal{C}$ is basis of $\mathcal{T}$.

b) For any $p \in X$, $U \in \mathcal{T}$ with $p \in U$, there exists a $U' \in \mathcal{C}$ such that $p \in U' \subseteq U$.

Proof.
$a) \implies b)$.
Let $p \in X$, and $U \in \mathcal{T}$ with $p \in U$ are given.
Since $a)$, $U = \bigcup_{U_\alpha \in A} U_\alpha$ for some $A \subseteq \mathcal{C}$. Now, for $x \in U$, there exists $U_\alpha \in A \subseteq \mathcal{C}$ such that $x \in U_\alpha \subseteq U$.
$b) \implies a)$.
Let $U \in \mathcal{T}$ be given. Then, by assumption, for any $x \in U$, there exists an open set $U_x \in \mathcal{T}$ such that

$$x \in U_x \subseteq U$$

Now,

$$U = \bigcup_{x \in U} U_x$$

is represented as the union of elements in $\mathcal{C}$. Therefore, $\mathcal{C}$ is a basis.

† Rigorous proof of $b) \implies a)$ using the Axiom of Choice)
Let $x \in X,\ U \in \mathcal{T}$ be given with $x \in U$ and $U \in \mathcal{T}$.
By assumption, for any $x \in X$, there is $U_x \in \mathcal{C}$ such that $x \in U_x \subseteq U$.
Now, for each $x \in X$, define non-empty set

$$S_x \overset{\text{def}}{=} \{V \in \mathcal{C} \mid x \in V \subseteq U\} \neq \emptyset$$

Let $\mathcal{S} \overset{\text{def}}{=} \{S_x \mid x \in U\}$. Then, since **Axiom of Choice**, there exists a choice fucntion

$$f : \mathcal{S} \to \bigcup_{x \in U} S_x \ \text{ s.t } \ f(S_x) \in S_x$$

That is,

$$f(S_x) = U_x \in \mathcal{C} \ \text{ s.t } \ x \in U_x \subseteq U$$

Now, construct

$$U = \bigcup_{x \in U} f(S_x) = \bigcup_{x \in U} U_x$$

□

0.2.1 Basic Theorems for Basis

> **Theorem 0.2.3.** Let X be a set, and $\beta \subseteq \mathcal{P}(X)$ be given.
> β is basis of for some topology of X if and only if it satisfies the following conditions:
>
> 1. For any $x \in X$, there exists a $B \in \beta$ such that $x \in B$.
> 2. For any $B_1, B_2 \in \beta$, $x \in B_1 \cap B_2$, there exists a $B \in \beta$ such that $x \in B \subseteq B_1 \cap B_2$.

Proof. If β is a basis, it trivially satiesfies above conditions. Suppose β is a collection satisfying these conditions. Define a *Topology on X generated by β* as:

$$\mathcal{T}_\beta \overset{\text{def}}{=} \left\{ \bigcup_{U \in A} U \;\middle|\; A \subseteq \beta \right\} \overset{(*)}{=} \{U \mid \forall x \in U, \ \exists B \in \beta \ \text{ s.t } \ x \in B \subseteq U\}$$

Now, enough to show that: $\mathcal{T}_\beta$ satisfies the Axiom of Topology. **A1, A2** are clear, To show **A3**:
Let $U_1, U_2 \in \mathcal{T}_\beta$ be given. Then, for any $x \in U_1 \cap U_2$, there exist $B_1, B_2 \in \beta$ such that

$$x \in B_1 \subseteq U_1, \ x \in B_2 \subseteq U_2$$

By assumption, there exists a $B \in \beta$ such that $B_x \subseteq B_1 \cup B_2$ with $x \in B_x$. Now,

$$U_1 \cap U_2 = \bigcup_{x \in U_1 \cap U_2} B_x$$

is contained in $\mathcal{T}_\beta$. □

> **Theorem 0.2.4.** Let β be a basis of a set X. Then,
>
> $$\mathcal{T}_\beta = \bigcap \{\mathcal{T} \subset \mathcal{P}(X) \mid \beta \subset \mathcal{T}, \ \mathcal{T} \text{ is topology}\}$$
>
> where $\mathcal{T}_\beta$ is the Topology generated by β.

Proof. Let $U \in \mathcal{T}_\beta$ be given. Then, for some $\mathcal{O} \subset \beta$, $U = \bigcup \mathcal{O}$.
Since the Topologies containing β also contains all elements of $\mathcal{O}$,
and there are topologies, so all topologies contain the union of $\mathcal{O}$. Thus U is contained in the intersection.
The converse is trivial, because $\mathcal{T}_\beta$ is a Topology containging β. □

0.3 Subbasis

Definition 0.3.1. Let X be a set. Define $\mathcal{S} \subseteq \mathcal{P}(X)$ is **subbasis** for a topology on X if:

$$X = \bigcup_{S \in \mathcal{S}} S$$

That is, for any $x \in X$, there exists a $S \in \mathcal{S}$ such that $x \in S \in \mathcal{S}$.
Define a **Topology generated by the subbasis** $\mathcal{S}$ as Topology by genereted the basis:

$$\beta_{\mathcal{S}} \stackrel{\text{def}}{=} \left\{ \bigcap_{S \in \mathcal{F}} S \;\middle|\; \mathcal{F} \subseteq \mathcal{S},\ \mathcal{F} \text{ is any finite set} \right\}$$

Clearly, $\mathcal{S} \subseteq \beta_{\mathcal{S}} \subseteq \mathcal{T}_{\beta_{\mathcal{S}}} = \mathcal{T}_{\mathcal{S}}$.

Lemma 0.3.2. Let X be a set, and $\mathcal{S} \subseteq \mathcal{P}(X)$ be a subbasis of X.
Then, $\mathcal{T}_{\mathcal{S}}$ is the smallest Topology on X containing $\mathcal{S}$.

Proof. Let $U \in \mathcal{T}_{\mathcal{S}}$ be given. Then, there exists a $\beta^* \subseteq \beta_{\mathcal{S}}$ such that

$$U = \bigcup_{B \in \beta^*} B.$$

For each $B \in \beta^*$, we have $B = \bigcap_{i=1}^{N} S_i$, where each S_i is an element of the subbasis $\mathcal{S}$.
Meanwhile, if $\mathcal{T}$ is a topology such that $\mathcal{S} \subseteq \mathcal{T}$, then each $S_1, \ldots, S_N$ is open in $\mathcal{T}$. Thus, the finite intersection

$$B = \bigcap_{i=1}^{N} S_i \in \mathcal{T}$$

Now, since U is the union of open sets in $\mathcal{T}$, it follows that $U \in \mathcal{T}$. □

Theorem 0.3.3. Let $\mathcal{S}$ be a subbasis of a set X. Then,

$$\mathcal{T}_{\mathcal{S}} = \bigcap \{ \mathcal{T} \subset \mathcal{P}(X) \mid \mathcal{S} \subset \mathcal{T},\ \mathcal{T} \text{ is topology} \}$$

where $\mathcal{T}_{\mathcal{S}}$ is the Topology generated by the subbasis $\mathcal{S}$.

Proof. Let $U \in \mathcal{T}_{\mathcal{S}}$ be given. Then, for some $\beta^* \subset \beta_{\mathcal{S}}$,

$$U = \bigcup_{B \in \beta^*} B = \bigcup_{B \in \beta^*} \bigcap_{S \in \mathcal{F}_B} S$$

where $\mathcal{F}_B$ is some finite subset of $\mathcal{S}$.
Since the topologies are closed under the finite intersection, the intersection B is contained all the topologies, and the topoloiges are closed under the union, the unions are contained all topologies. The converse is trivial. □

0.4 Comparison of Topologies

Theorem 0.4.1. X set, and $\mathcal{T}_1, \mathcal{T}_2$ are Topology. TFAE:

a) $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

b) For any $U \in \mathcal{T}_1$, $x \in U$, there exists $V_x \in \mathcal{T}_2$ such that $x \in V_x \subseteq U$.

Theorem 0.4.2. Let X be a set, and β_1, β_2 be basis of X. TFAE:

a) $\mathcal{T}_{\beta_1} \subseteq \mathcal{T}_{\beta_2}$.

b) For any $B \in \beta_1$, $x \in B_1$, there exists $B_2 \in \beta_2$ such that $x \in B_2 \subseteq B_1$.